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New African Lower Carboniferous paleomagnetic pole from intrusive rocks of the Tin Serririne basin (Southern border of the Hoggar, Algeria)

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Abstract

A paleomagnetic study has been conducted on intrusive doleritic rocks cropping out within Devonian horizontal tabular formations of the Saharan craton (Tin Serririne basin, South of Hoggar shield). The $^{40}K/^{40}Ar$ dating of the dolerites gave an age of 347.6±8.1 Ma, i.e. Tournaisian. The paleomagnetic data present three different directions. The first has a paleomagnetic pole close to the previous African poles of Permian age. This direction is therefore interpreted as a Permian remagnetization. The second direction, which is defined by both linear regression and remagnetization circles analysis, is considered as the primary magnetization. It yields a new African Tournaisian paleomagnetic pole (λ=18.8° S, ϕ=31.2° E, K=29, A_{95}=7.5°) very close to the Ben Zireg Tounaisian pole [Aifa, T., Feinberg, H., Pozzi, J.P., 1990. Devonian/Carboniferous paleopoles for Africa. Consequences for Hercynian geodynamics. Tectonophysics, 179, 288–304]. The third direction has intermediate orientation between those of the first or second directions and that of the Upper Cenozoic field. It is interpreted as related to a composite magnetization. This new Tin Serririne pole improves the APWP of Gondwana, for this key period of the evolution of the Pangea. This APWP confirms the previous paleogeographic reconstruction which shows that the pre-Hercynian ocean between Gondwana and Laurussia is still not close during the beginning of the Carboniferous.
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Keywords: Apparent Polar Wander Path (APWP); Lower Carboniferous; Gondwana; Sahara craton; Dolerites

1. Introduction

The formation of the Pangea supercontinent by the collision of Gondwana and Laurussia was one of the most important geological events during the Upper Paleozoic times. However, the detail of the convergence

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between these two plates, before collision, is still of debate, e.g. their relative position during the Upper Devonian–Lower Carboniferous times and the question, whether the ocean separating Gondwana from Laurussia was wide or not at that time, remains controversial. These uncertainties are due to the lack of reliable paleomagnetic data from the Gondwana continents for this key period, leading to different Apparent Polar Wander Paths (APWP) for this supercontinent (e.g. Morel and Irving, 1978; Schmidt et al., 1990; Bachtadse and Briden, 1990; Van der Voo, 1993; McElhinny et al., 2003). The African APWP for the Paleozoic times is not very well defined, especially for the Upper Devonian–Lower Carboniferous (Bachtadse et al., 1987; Aifa et al., 1990; Bachtadse and Briden, 1991; Smith et al., 1994). Several paleomagnetic studies carried out on Devonian and Carboniferous sedimentary rocks in the stable Saharan craton (southern Algeria) in order to improve this curve, but the results correspond rather to secondary magnetizations (Aifa, 1993; Derder et al., 1997; Bayou et al., 2000; Henry et al., 2004a).

Because magmatic rocks often present more accurate magnetic characteristics than those obtained from sedimentary rocks, paleomagnetic investigation was carried out on doleritic rocks. The dolerites subject to this study intruded the horizontal tabular Devonian formations of the Tin Serririne basin during Lower Carboniferous times.

2. Geological setting

The Hoggar shield, also called "Ahaggar", consists of mesozonal to catazonal series of Archean, Eburnean and Neoproterozoic lithologies that are later intruded by granitic rocks. The Tin Serririne–Tin Mersoï basin is located on the south border of the Hoggar shield and extends southeasterly covering the northern part of Niger (Fig. 1). It is composed of Paleozoic series that overly the Proterozoic basement of the Hoggar shield. Cambrian magmatic and metasedimentary complexes form the base of the series and outcrop at the In Guezzam region (Djellit et al., 2002). The overlying sedimentary rocks are of Ordovician to Carboniferous age in Algeria and extend to Permian in Niger.

The synclinal structure of Tin Serririne (Fig. 1) constitutes a large depression limited to the west by the In Guezzam fault system (Djellit et al., 2002). The syncline is asymmetric with a relatively flat bottom that becomes irregular at the fault system zone of In Guezzam. It is also crossed to the east by major N–S Late PanAfrican lineament called the 7° 30 shear zone (Fig. 2) (Black et al., 1994; Liégeois et al., 2003; Henry et al., 2004b). Mafic rocks are locally exposed at both sides of the syncline but are most widespread at the eastern border, just near the major accident (Figs. 1 and 2). They are interbedded within the horizontal Lower–Middle Devonian formation (Djellit et al., submitted for publication) which corresponds mainly to successive sedimentary cycles (Fig. 2) of coarse to fine sandstone. The contacts between magmatic rocks and host rocks cannot be observed, due to very bad outcrop conditions. Bournas (1998) indicates these rocks as basaltic flows in his stratigraphical description, but no evidence of flows has been observed. On the contrary, in a narrow outcrop, local symmetrical tilting around a NW–SE direction can be observed in series overlying the magmatic rocks, indicating probably that the latter form here a thin dyke. Moreover, levels overlying magmatic rocks are baked or affected by hydrothermal phenomenon, showing that the basic rocks were intruded within the sedimentary formations. Anisotropy of magnetic susceptibility measurements (Djellit et al., submitted for publication) confirmed the intrusive character of these rocks: subhorizontal magnetic foliation obtained in some sites indicates emplacement as sill but subvertical magnetic foliation on the contrary characterizes intrusion as dyke in other sites. Owing the very small difference in elevation of the host rocks under and above the sill, the latter should be thin. The fact that these sills and dykes are all located within the same level of the Lower–Middle Devonian is surprising, because this level has apparently similar sedimentary characteristics as the overlying levels. Field observations suggest that levels intruded by dolerites present more intense NW–SE fracturing and faulting than the overlying levels. That should indicate weak tectonic event during the Lower–Middle Devonian and could explain the fact that the dolerites were not found in the upper part of the Lower–Middle Devonian.

Thin section studies show that these rocks present doleritic texture. These dolerites are usually dark colored. Black varieties are unaltered, whereas the green ones are weathered. Microprobe analyses indicate that they are composed of An29–An56 plagioclases, pyroxenes of Wo46 to Wo48, En24 to En32 and Fs20 to Fs27, opaque minerals and apatite. The plagioclases laths are intergrowth and pyroxenes are interstitial. The most common visible accessory minerals are opaque oxides (7–8%) which occur as granules showing either squeletal textures or subhedral crystals. They are represented by ilmenite and other Ti-rich oxides. Apatite often crystallizes as fine elongated crystals and is ubiquitous in the rock.

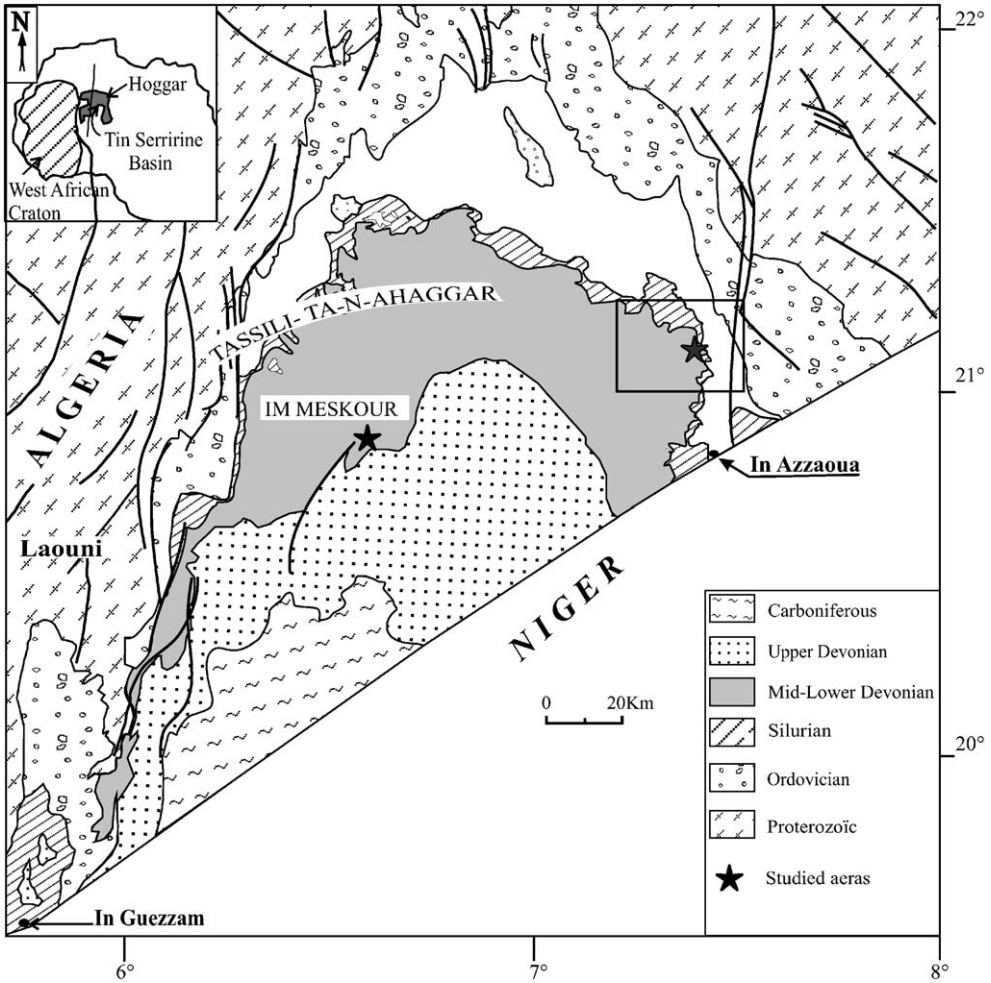


Fig. 1. Geological map of the studied area with location of the dolerites outcrops. The outcrops area on the western part of the basin (Im-Meskour area) is of about 1 km^2.

These rocks are identical to those previously mentioned west of the In Guezzam by Lessard (1962) who attributes their emplacement to the Hercynian times. In practice, for these dolerites, no arguments allow to specify the emplacement age, which could be Lower Devonian or more recent. To have usable paleomagnetic data, absolute dating is key information. The isotopic age results (Djellit et al., submitted for publication) are listed in Table 1. They range between 341.8 and 352.6 Ma and the resulting mean age is 347.6 ± 8.1 Ma which corresponds to a Tournaisian age (Odin, 1994; Gradstein et al., 2004).

3. Sampling and analysis procedures

In the Tin Serririne syncline, the sampling sites were located both in the eastern part (21° 06′ N, 7° 23′ E) in the In Azzaoua area (Figs. 1 and 2), near the border between Algeria and Niger (Fig. 2) and in the western part (20° 48′ N, 6° 28′ E) in the Im-Meskour area (Fig. 1). 291 oriented cores of unaltered dolerites distributed over 33 sites have been collected. Two limited outcrops of Lower–Middle Devonian sandstone, one baked close to the dolerites and another one farther from these rocks, were also sampled in order to perform a contact test.

One to three specimens of standard size (cylinders of 11 cm^3) were cut from each sample, mostly allowing both demagnetization treatments and additional rock magnetic studies to be performed. Prior to any demagnetization analysis, the specimens were stored in zero magnetic field for at least 1 month in order to reduce a possible viscous magnetization. The Remanent Magnetization was measured using JR5 spinner magnetometer (AGICO, Brno, Czech Republic).

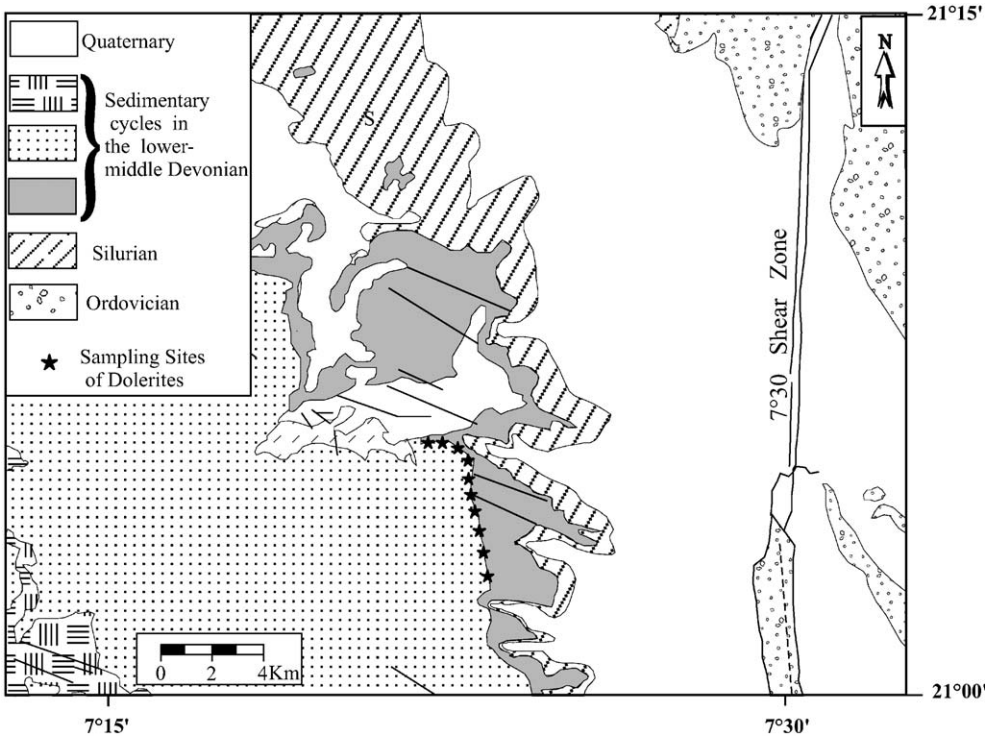


Fig. 2. Geological map of the In Azzaoua area and location of the sampling sites.

Half of the cores distributed over 12 sites were cut by a new acquired saw (ASC). Unfortunately, it has been proved later that the blades of this new saw were magnetized, giving strong parasitic magnetization, which cannot be separated from the NRM components during demagnetization. So, the paleomagnetic results obtained from these samples were discarded.

In order to correctly isolate and identify the magnetization components, numerous demagnetization steps were performed with increments ranging from 100 °C and less in the lowest temperatures to 10 °C in the highest ones, whereas the increments range from 2.5 mT to 10 mT during the Alternating Field (AF) demagnetization (maximum used field intensity: 90 mT).

The results of demagnetization process are presented on orthogonal vector plots (Wilson and Everitt, 1963; Zijderveld, 1967). The vectors remaining after each step and the difference vectors removed between two consecutive demagnetization steps were plotted on equal area projections. When clearly identified, the

Table 1
^{40}K–^{40}Ar dating of the doleritic sill

Sample	Analysis number	Mean age	Age	Standard deviation Ma	$^{40}Ar^{*}$ (%)	$^{40}Ar^{*}/g$ (10^{-7} cm^3/g)	K_2O (%)	Weight (g)
D3 WR	6731	345±8 Ma	341.8	7.8	95.8	178.3	1.47	0.3065
	6734		347.5	8	95. 2	181.6		0.3007
	6751		347.5	8	96.8	181.6		0.4025
	6772		343.4	7.9	96.4	179.3		0.4033
D1 WR	6708	347.6±8 Ma	342.8	7.9	97.8	174.1	1.43	0.5059
	6743		351.6	8	97	178.9		0.4044
	6750		351.6	8	96.3	178.9		0.4000
	6771		344.4	7.9	96.6	175.3		0.4073
DH WR	6752		350.3	8	97	184.5	1.48	0.4015
DH Fds	6754		352.6	8.1	95.4	250	1.99	0.1386

Ages were calculated using the constants recommended by Steiger and Jäger (1977) and the errors at one sigma level are quoted following proposals by Mahood and Drake (1982). Mean age WR (whole rock fraction) and Fds (feldspar grains) for the sample DH: 351.4 Ma. Mean age WR for the samples D3, D1 and DH: 347.6 Ma (after Djellit et al., submitted for publication).

direction of the magnetization components was calculated by principal component analysis (Kirschvink, 1980), otherwise remagnetization circles (Halls, 1976; Halls, 1978; McFadden and McElhinny, 1988) were used when two adjoining magnetization components had partly or totally overlapping unblocking temperatures (unblocking fields) spectra. Mean characteristic directions were calculated using Fisher's statistics (Fisher, 1953).

4. Rock magnetism analysis

Thermomagnetic curves (low-field susceptibility as a function of temperature) were determined in argon and air using CS3 equipment—Kappabridge KLY3 (AGICO, Brno, Czech Republic). They show a first sharp decrease of susceptibility at about 300° to 400 °C according to the samples and another around 550–580 °C (Fig. 3a). The curves are reversible until slightly higher temperatures than the first sharp decrease. This decrease corresponds then to a Curie temperature. This Curie temperature is higher than that of pyrrhotite. Moreover, no sulfides have been observed in thin sections and from microprobe analysis. On the contrary Ti-rich oxides have been found and this Curie temperature is probably related to these oxides. After heating at temperatures higher than about 400 °C, the thermomagnetic curves become irreversible. Mostly on the cooling curve, there is disappearance of the Curie temperature at around 340° to 400 °C and occurrence of a new Curie temperature at lower temperature (Fig. 3b). The interpretation of this behavior is quite difficult and disputable (Henry et al., submitted for publication). The thermal instability of the main magnetic component suggests that this component is titanomaghemite. But the observed progressive transformation during subsequent thermal treatments at increasing maximum temperature is not simply inversion to titanohematite. Such inversion, as well as transformation into titanomagnetite with the same Ti-content, cannot explain the observed progressive decrease of both the Curie temperatures and of the susceptibility. Simple progressive exsolution should have produced another strongly magnetic component with higher Curie temperature. The alteration mechanisms are probably more complex: for example, progressive exsolution producing Ti-enriched titanomaghemite and Ti-poor maghemite which inverted to hematite. Another possibility could be an effect of small amount of magnesioferrite (see Henry et al., submitted for publication). The second sharp decrease of susceptibility around 550–580 °C

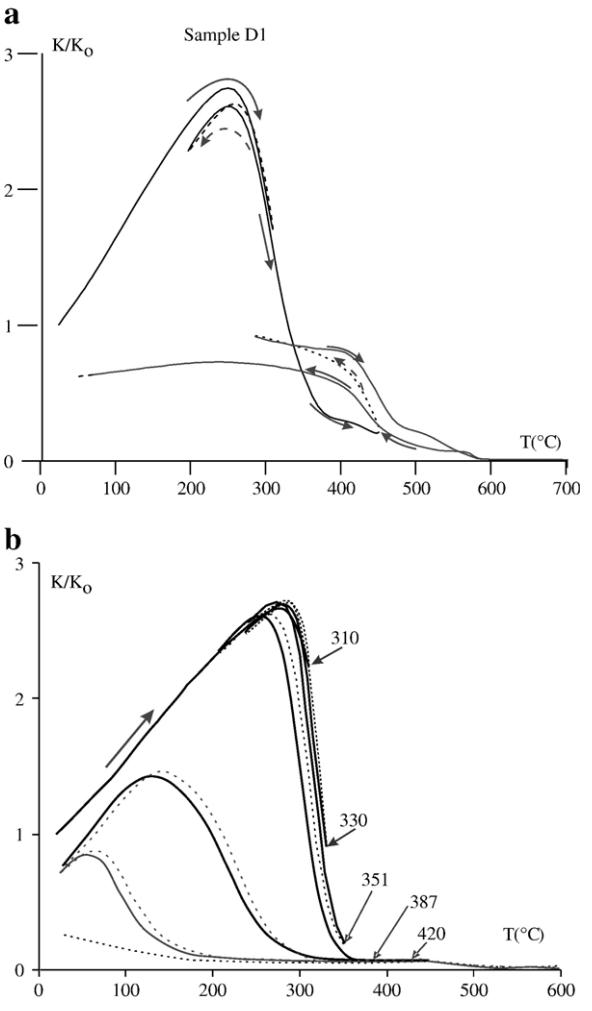


Fig. 3. (a) Thermomagnetic curve of sample D1 (susceptibility in low field as a function of temperature) pointing out a Curie point at 340 °C and another around 580 °C, and mineralogical alteration at temperature higher than 340 °C (dotted lines: partial cooling curves). (b) Typical thermomagnetic curve of sample D133, showing on partial loops variation of the Curie temperature from 310 °C to lower temperatures.

corresponds to Curie temperature of magnetite, but, because of occurrence of mineralogical alteration at lower temperature, this magnetite could be mineral pre-existing in the rock as result of this alteration. However, the maximum blocking temperatures obtained on part of the samples from thermal demagnetization of the Natural Remanent Magnetization (NRM) are around 580 °C, indicating the presence of magnetite (Fig. 4). This magnetite is then pre-existing. It has to be noticed that few samples show maximum blocking temperature higher than 630 °C, evidencing existence also of hematite (that clearly appears only in one site entirely discarded because of

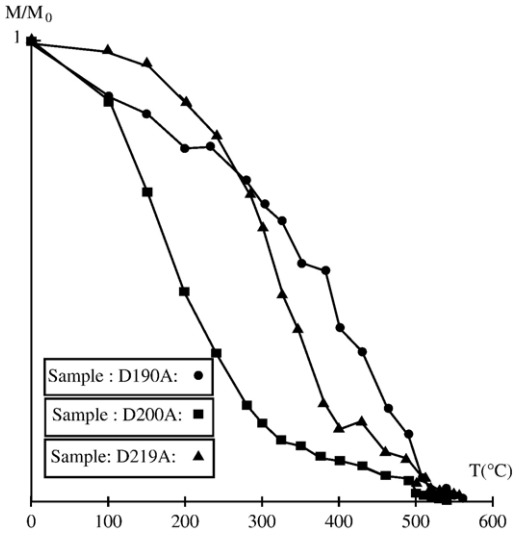


Fig. 4. Examples of Natural Remanent Magnetization (NRM) thermal demagnetizations curves showing maximum blocking temperatures around 550–580 °C.

remagnetization due to the magnetic saw, but suggests that small amount of hematite could be present in the other sites).

Main ferrimagnetic minerals of these dolerites are probably Ti-rich titanomaghemite and magnetite, but hematite is also clearly present in some samples. On the Curie curves, sharp decrease of susceptibility corresponding to the titanomaghemite is much stronger than that associated to magnetite, and in these rocks magnetite content is probably lower than titanomaghemite content. Magnetite was not found in the large grains studied by microprobe analysis and probably corresponds to very small grains.

Hysteresis loops were obtained for small cores (cylindrical samples of $1 cm^3$) using a laboratory-made translation inductometer within an electromagnet capable of reaching 1.6 T (Fig. 5). The coercive force deduced from several representative samples is low (lower than 10 mT), and the saturation of the magnetization is complete at approximately 0.4 T pleading for the presence of mineral phase of magnetite type. Hcr/Hc is 2.2 whereas Jrs/Js is 0.08, but we cannot infer information about grain size because of the presence of different magnetic minerals.

5. Paleomagnetic results

The magnetic behavior during demagnetization observed in the eastern part of the basin (In Azzaoua area) is very similar to that from the western part (Im-Meskour area). Thermal or AF demagnetizations were performed on all samples. Because of the strong mineralogical changes during the heating, AF demagnetization gets better results than the thermal one. During the demagnetization process, the magnetization direction of samples evolves mostly along great circles (remagnetization circles), which shows superimposition of the unblocking temperature (or unblocking fields) spectra for at least two remanent magnetization components (Fig. 6). During thermal demagnetization, the overlapping of unblocking temperature spectra is limited to temperature lower than the Curie temperature of the titanomaghemite. It is therefore due to the presence of both titanomaghemite and magnetite. However, most of the samples do not allow defining a direction after demagnetization of the titanomaghemite because their remanent magnetization carried by magnetite has too weak intensity. Similar observation can be made for AF demagnetization. For a few samples, part of the evolution on great circles corresponds to linear segments on Zijderveld plots, and a direction can be defined by principal component analysis (Kirschvink, 1980). Most of these directions, which do not correspond to Characteristic Remanent Magnetization (ChRM), are scattered. As the corresponding NRM intensities of these samples are very high (up to $2.3\ 10^{-2}$ A/m), this scattering is probably due to lightening effects. In the other samples with sufficient magnetization carried by magnetite, a ChRM has been obtained at high blocking temperature until maximum temperature of 580 °C (or relatively high blocking fields 90 mT sometimes). This ChRM presents also scattered direction in some samples, which is probably due to lightening too.

In the In Azzaoua area, the non-scattered ChRMs allow to define two neighboring clusters of directions:

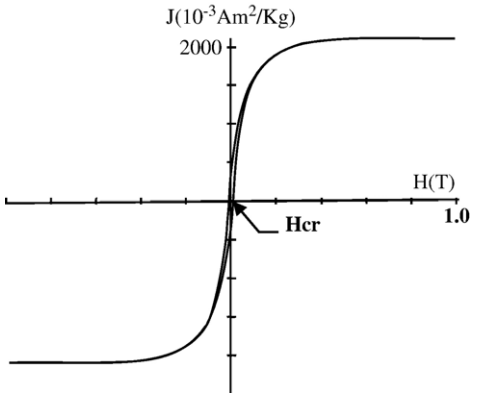


Fig. 5. Hysteresis loop of a representative sample, showing low coercive force.

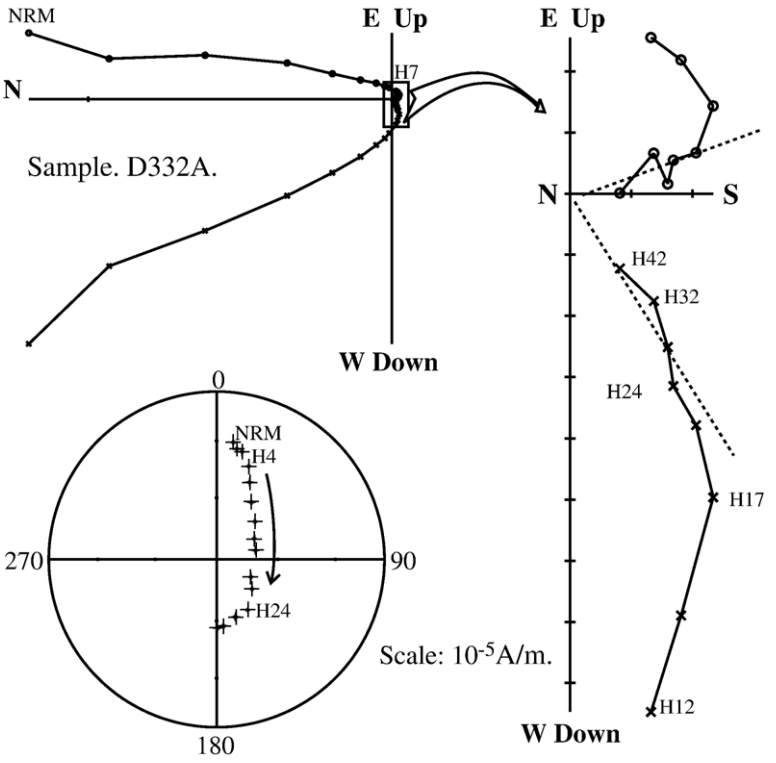


Fig. 6. Equal area plot (lower hemisphere) and orthogonal vector plots (open circles: horizontal plane, crosses: vertical plane) for specimen D332A, showing evolution along great circle during the demagnetization of the NRM, but reaching a stable orientation at high blocking fields (H: demagnetization field intensity in mT).

the ChRM A (Fig. 7) has been isolated from 7 sites (N=13 samples, D=142.3°, I=32.9°, k=68, α_{95}=5.1°). The ChRM B (Fig. 8) was obtained on 4 sites (N=11 samples, D=158.5°, I=62.3°, k=37, α_{95}=7.6°; Table 2). Thirty-one samples yield only well defined great

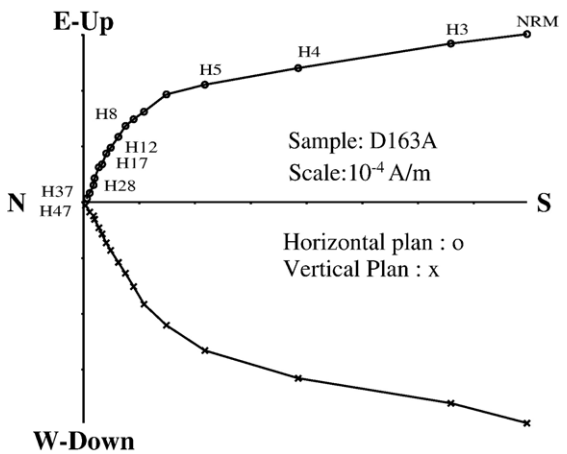


Fig. 7. Orthogonal vector plots for sample displaying the ChRM A (open circles: horizontal plane, crosses: vertical plane, H: demagnetization field intensity in mT).

circles. Their best intersection direction using Halls (1978) method is D=142.0°, I=60.2°. It is close to the direction B. Combining the ChRM B and the remagnetization circles data using McFadden and McElhinny (1988) method (each sample was used obviously once, either as ChRM or circle), the direction B is then D=147.7°, I=60.2°, k=49, α_{95}=3.1° (Table 2). The directions, which do not correspond to ChRM, also present a dominant orientation, allowing definition of a direction C (N=9 samples, n=4 sites, D=173.6°, I=28.1°, k=32, α_{95}=9.2°) different from that of the ChRMs A and B (Fig. 9).

Results obtained in the western part of the basin (Im-Meskour area) are very similar. The ChRM A was found in 4 sites (N=samples, D=0.6°, I=39.3.1°, k=39, α_{95}=9.0°) and the ChRM B in 4 sites (N=13 samples, D=147.6°, I=63.9°, k=54, α_{95}=5.7°).

Similarly, 18 samples yield only remagnetization circles and give a direction of best intersection of great circles very close to that obtained in the In Azzaoua area: D=143.1°, I=65.2°. Combining the ChRM B and the remagnetization circles data using McFadden and McElhinny (1988) method, the obtained direction is D=145.9°, I=64.2°, k=55, α_{95}=3.6° (Table 2). The

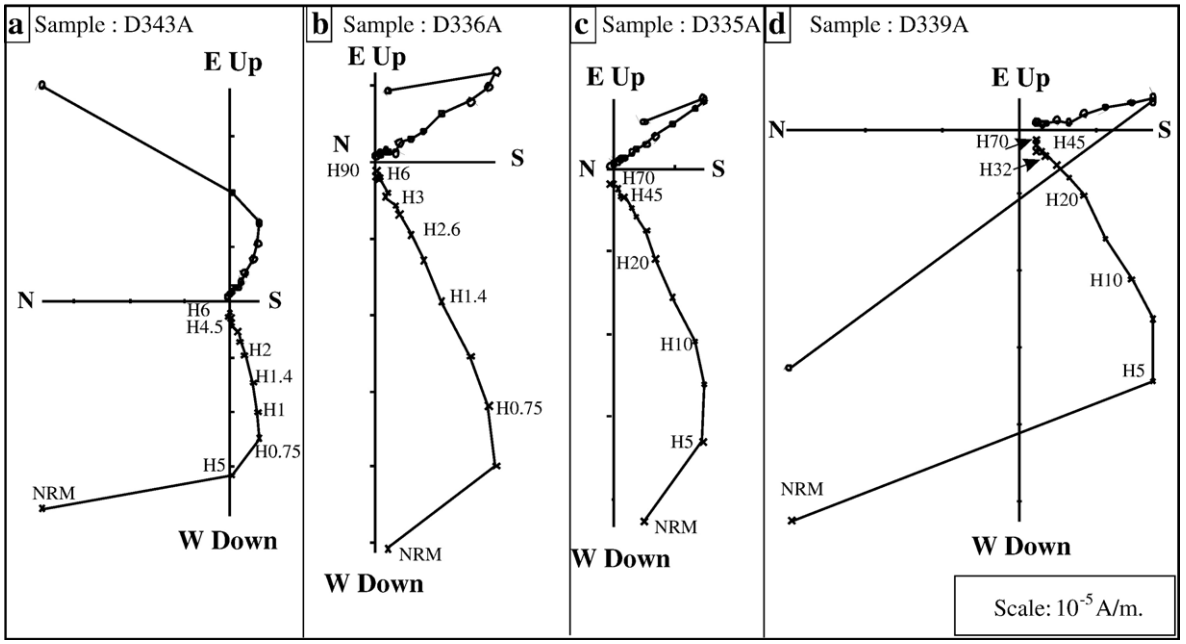


Fig. 8. Orthogonal vector plots for samples displaying the ChRM B (open circles: horizontal plane, crosses: vertical plane, H: demagnetization field intensity in mT).

component C is also present in 9 sites (N=14 samples, D=168.5°, I=33.8°, k=34, α_{95}=6.6°).

The paleomagnetic data obtained in the two parts of the Tin Serririne syncline are identical for ChRMs A and B as for the component C. They can therefore be merged. Here, intrusion occurred in horizontal formation and no structural correction has to be applied.

- ChRM A (Fig. 10): n=11 sites is D=136.7°, I=35.4°, k=39, α_{95}=5.2°, paleomagnetic pole at λ=36.3°S, ϕ=53.6°E.
- ChRM B (Figs. 11 and 12; Table 2), combined with great circles: n=12 sites, D=148.1°, I=61.4°, k=48, α_{95}=5.9° (N=73 samples, D=145.0°, I=63.0°, k=54, α_{95}=2.3°), paleomagnetic pole at λ=18.8° S, ϕ=31.2° E.
- Direction C (Fig. 13): n=13 sites: D=169.7°, I=32.3°, k=35, α_{95}=5.2°, paleomagnetic pole at λ=50.2° S, ϕ=22.4° E.

The "ellipticity test", introduced by Henry and Le Goff (1994) and based on the bivariate statistics (Le Goff, 1990; Le Goff et al., 1992), allows to point out non-Fisherian distribution of paleomagnetic directions. Contrary to ChRMs A and B, the component C presents a significantly elliptical distribution, as the ky/kx ratio of 3.29 is largely higher than the critical value for 23 samples (2.06). Its calculated mean direction is, however, the same using Fisher and bivariate statistics.

Unfortunately, the samples from the baked and unbaked sandstone have only very low and unstable remanent magnetization and it was then not possible to perform a contact test.

6. Discussion

6.1. Paleomagnetic results of the Tin Serririne basin

The ChRM A corresponds to a paleomagnetic pole close to the previous African poles of Permian remagnetizations (Daly and Irving, 1983; Aifa, 1987; Henry et al., 1992, 2004a; Derder et al., 2001c). It is interpreted therefore as a remagnetization acquired during Permian times.

The direction of the ChRM B is different from that of the well known remagnetization directions in the Saharan craton (Bayou et al., 2000; Henry et al., 2004a). It is also different from that associated to poles since Upper Carboniferous–Permian for Africa (Morel et al., 1981; Daly and Irving, 1983; Henry et al., 1992, 1999; Merabet et al., 1998, 1999; Derder et al., 2001a,b, c). Though no paleomagnetic tests were possible here, this ChRM B is then interpreted as the primary magnetization defined by n=12 sites, D=148.1°, I=61.4°, k=48, α_{95}=5.9° (Table 2). The corresponding

Table 2
Mean direction (Declination D and Inclination I) of the ChRM B in each site, calculated using combination of great circles and ChRM data, and their corresponding paleomagnetic poles (λ and ϕ are the latitude and longitude of the pole, respectively), k, α_{95}, K and A_{95} are the corresponding Fisher's parameters

Sites	N (circles)	N ChRM	D (°)	I (°)	k	α_{95} (°)	λ (°S)	ϕ (°E)	K	A_{95} (°)
In Azzaoua area (21.1° N, 7.38°E)										
19+20	1+5	–	140.0	47.4			27.1	46.8		
23	7	2	150.2	57.4	133	4.8	24.6	32.9		
21+22	3	1	147.3	77.5	70	15.1	−0.6	20.0		
25	3	–	178.8	45.6			41.8	8.8		
26	3	1	150.4	64.0	234	8.2	18.0	28.6		
43+45	7+2	7	149.4	61.3	34	6.5	20.5	31.0		
44	4	–	144.9	48.9			28.9	42.1		
Mean samples	31	11	147.7	60.2	49	3.1	21.0	32.9		
Mean 7 sites			154.8	60.3	40	8.3	23.4	27.8		
Mean sites							23.4	30.3	23	11.1
Im-Meskour area (20.8°N, 6.46°E)										
51		7	153.8	63.4	94	6.3	20.0	25.9		
52	3	4	136.3	66.2	58	8.4	10.4	34.1		
55	3	1	121.7	63.5	16	16	5.5	43.6		
58	3	1	150.5	62.7	63	15.8	19.7	28.5		
60	3		139.6	70.5			7.0	28.6		
Mean samples	18	13	145.9	64.2	55	3.6	16.4	30.4		
Mean 5 sites			140.6	65.7	163	4.9	12.7	32.3	60	9.0
Mean 5 sites							12.6	32.3	69	7.6
All sites										
Total samples	49	24	145.0	63.0	54	2.3				
Total 12 sites			148.1	61.4	48	5.9				
Total 12 sites							18.8	31.2	29	7.5

paleomagnetic pole (λ=18.8° S, ϕ=31.2° E, K=29, A_{95}=7.5°), located in the present south-central Africa, constitutes then a new African Lower Carboniferous paleomagnetic pole. It places the studied site area at a paleolatitude of about 42° S.

Concerning the component C, its elliptical distribution suggests that it is a composite magnetization, which components cannot be separated by thermal and AF treatments (Bouabdallah et al., 2003). In addition, the direction of this component is often included in a plane containing ChRM A or B and a direction which should have been that of the Upper Cenozoic remagnetizations

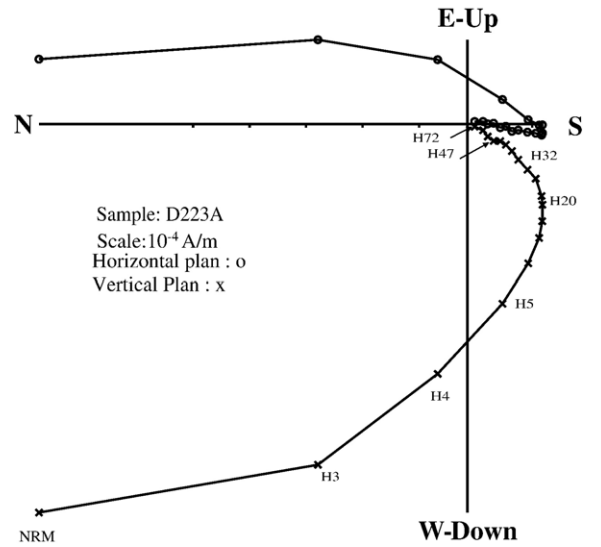


Fig. 9. Orthogonal vector plots for samples displaying the component C (open circles: horizontal plane, crosses: vertical plane, H: demagnetization field intensity in mT).

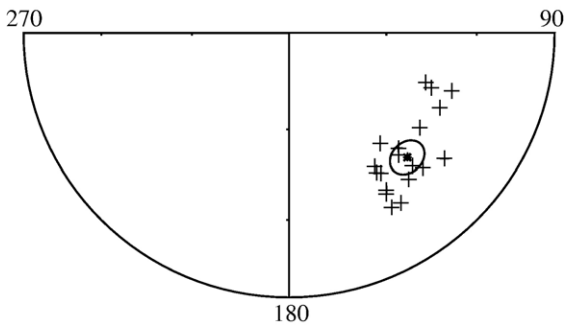


Fig. 10. Equal area plot (lower hemisphere) of the ChRM A directions; the star represents the mean direction and is associated to its 95% confidence cone.

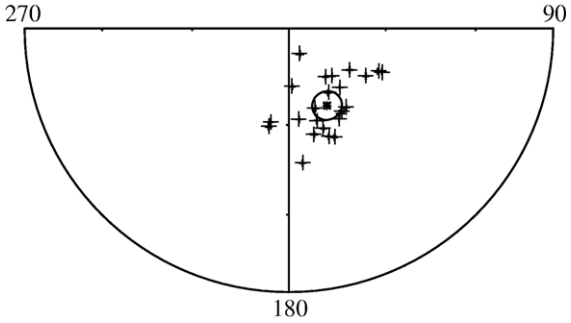


Fig. 11. Equal area plot (lower hemisphere) of ChRM B, see caption of Fig. 10.

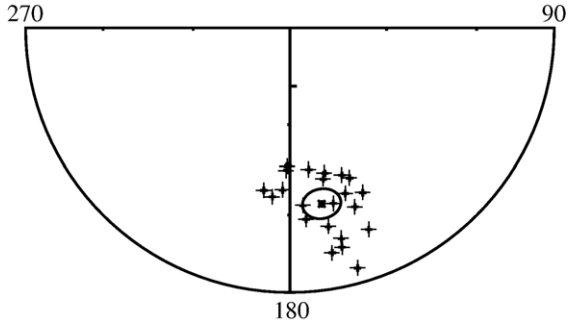


Fig. 13. Equal area plot (lower hemisphere) of the component C, see caption of Fig. 10.

(Fig. 14), with normal or reversed polarity, found in many formations in the Saharan platform (Derder et al., 1997, 2001a; Bayou et al., 2000; Henry et al., 2004a). The component C is in an intermediate position between the ChRM A or B and the direction of the Cenozoic field, with normal or reversed polarity. A good example for the composite character of the magnetization is given by the sample D190A. Its NRM has a direction close to that of the ChRM B. During demagnetization, this direction evolves towards that of a Cenozoic direction with reversed polarity but it yields a direction C for some demagnetization steps. However, for the last steps of the demagnetization treatment, the direction again slightly evolves towards the Cenozoic direction (Fig. 15). The composite character of such component can be also illustrated by the paleomagnetic results obtained from the site 53 (Fig. 16), where the direction obtained in the different samples is scattered in a great circle containing the direction C. We can notice that this great circle does not include Cenozoic direction and the second component is here unknown (lightening effect?). Similar composite components determined by principal component analysis (Kirschvink, 1980) have been already observed in the Saharan platform (Henry et al., 1999; Merabet et al., 1999; Derder et al., 2001a; Bouabdallah et al., 2003).

6.2. Apparent Polar Wander Path—APWP

For Gondwana, the Upper Devonian–Lower Carboniferous poles are highly scattered, and a controversy

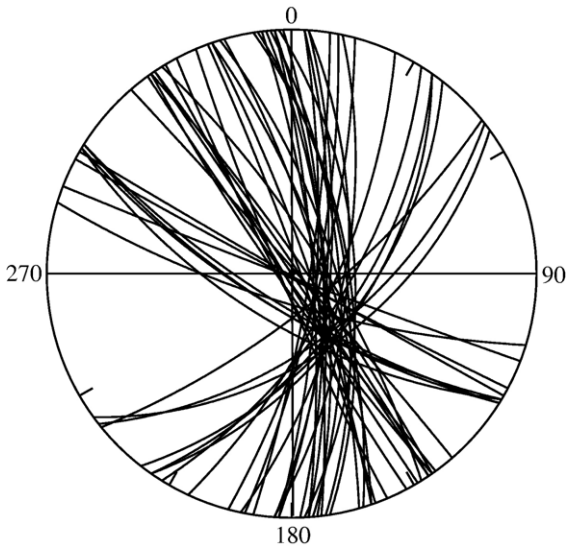


Fig. 12. Equal area plot (lower hemisphere) showing the best intersection direction obtained from the remagnetization circles analysis.

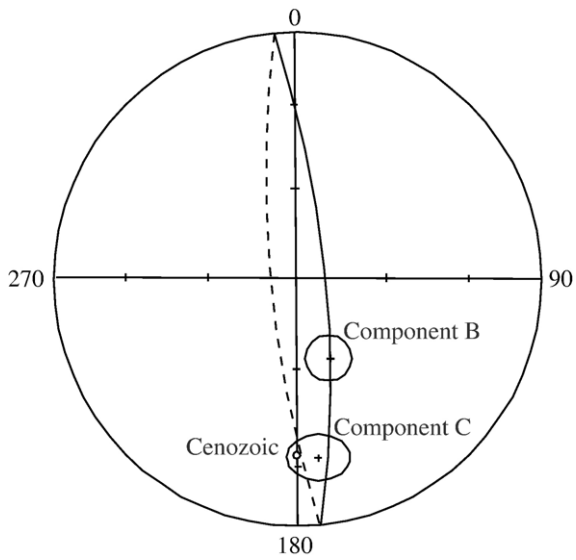


Fig. 14. Equal area plot showing that the component C is included in a plane containing ChRM B and a field direction during the Upper Cenozoic with reversed polarity: crosses and continuous line in the lower hemisphere, open circle and dotted line in the upper hemisphere.

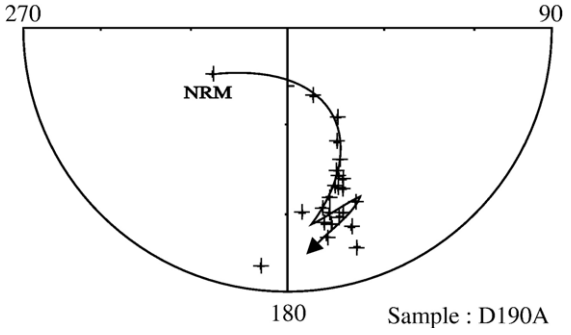


Fig. 15. Equal area plot (lower hemisphere) of sample D190A, suggesting a composite character for component C (see text).

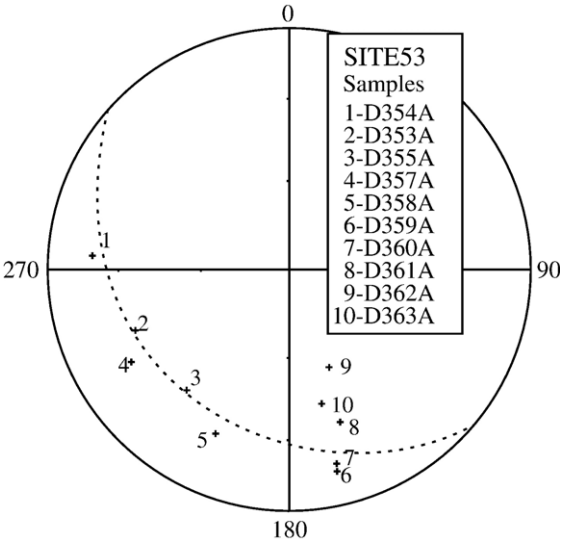


Fig. 16. Equal area plot (lower hemisphere) of the directions obtained in the different samples of site 53; these directions are scattered in a great circle containing the component C, pleading for its composite character.

still exists regarding its APWP (e.g. Morel and Irving, 1978; Schmidt et al., 1990; Bachtadse and Briden, 1990; Van der Voo, 1993; Smith et al., 1994; Chen et al., 1993; McElhinny and McFadden, 2000). This could be due to the discrepancy often observed between the APWPs of West and East Gondwana. However, very few reliable poles from the West Gondwana are available, whereas most of paleomagnetic poles of the East Gondwana are from Australia. Many of these Australian poles are derived from the Tasman Fold Belt in eastern Australia, where tectonic movements are suspected (Powell et al., 1990; Li et al., 1990). McElhinny et al. (2003) published a new compilation of the Gondwana Paleozoic poles, including poles from eastern Australia only from terranes accreted to cratonic Australia before formation of the studied rocks. They therefore proposed a new Gondwanian APWP.

In their compilation, McElhinny et al. (2003) subdivided the Paleozoic into periods of 20 Ma interval. Tin Serririne pole has age spanning two of these subdivided periods: the Lower Carboniferous (330–350 Ma) and the Late Devonian–Early Carboniferous (350– 370 Ma). When comparing our new datum with those of this compilation, it appears that our pole is not very far from the Gondwana mean pole for the Lower Carboniferous (λ=1.5°S, ϕ=41.7°E, K=24, A_{95}=14.4°). It is also not significantly different (partial overlapping of confidence zones) from mean pole for Late Devonian–Early Carboniferous period (λ=9.3°S, ϕ=9.7°E, K=44 and A_{95}=8.0°) of this compilation. Tin Serririne pole also agrees with the mean Upper Devonian–Middle Carboniferous pole (λ=21°S, ϕ=47°E) for West Gondwana given in the compilation of Van der Voo (1993). We can also notice that our new pole is close to the Famenian one (λ=19.2°S, ϕ=19.8°E) of Ben Zireg in Algeria and is very similar to that attributed to Tournaisian (λ=25.3°S, ϕ=21.1°E), in the same area, from remagnetization results (Aifa et al., 1990).

New data from the Saharan craton in Algeria (Table 3) were added to the McElhinny et al. (2003) paleomagnetic database in order to complete their proposed APWP curve. On the contrary, Griotte limestone pole has been discarded, because in fact, it corresponds to the paleomagnetic data from Ben Zireg recalculated for another location. A new APWP was determined from this selection by bivariate statistics (Le Goff, 1990; Le Goff et al., 1992). This approach uses all

Table 3
Data from Sahara craton (Algeria) added to the compilation of McElhinny et al. (2003)

Age	Location	λ (°S)	ϕ (°E)	K	A_{95} (°)	Reference
Serpukhovian–Moscovian	Ain Ech Chebbi	26.5	44.7	383	4.7°	Derder et al. (2001a)
Bashkirian	El Adeb Larache	28.2	55.5	207	3.4°	Derder et al. (2001b)
Moscovian	Edjeleh	28.3	58.9	157	4.2°	Derder et al. (2001c)
Namurian	Reouina	28.4	56.9	642	1.7	Merabet et al. (1999)
Stephanian	Merkala	32.4	56.6	399	2.3°	Henry et al. (1999)
Stephanian–Autunian	Tiguentourine	35.3	60.3	195	3.2	Derder et al. (1994)

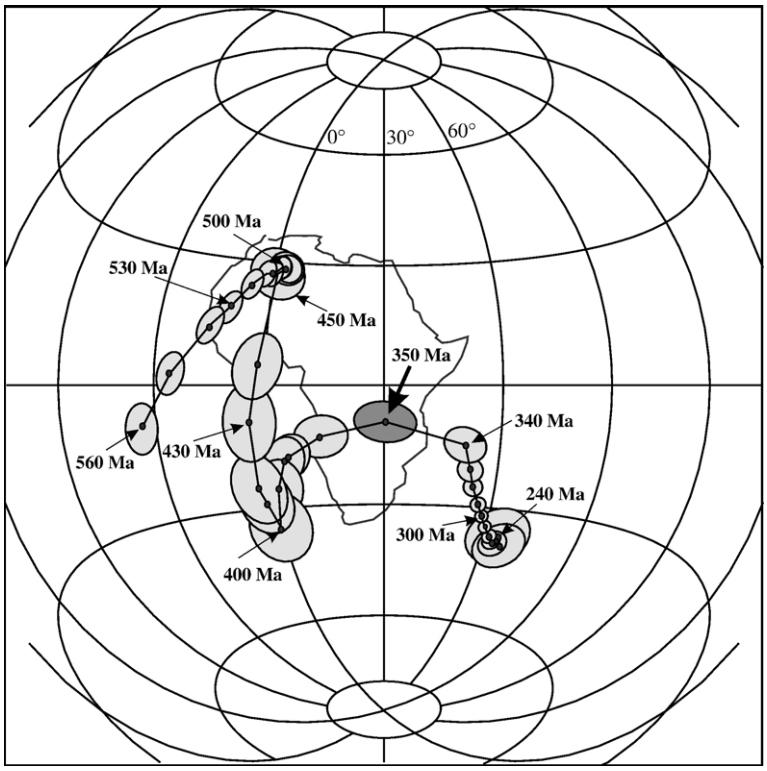


Fig. 17. APWP of Gondwana obtained using bivariate statistics (Le Goff, 1990; Le Goff et al., 1992) with "weighting time" method (Le Goff et al., 2002).

data having at least part of their uncertainty windows in times within a window of 20 My around the considered age. Each data is weighted by the part of its uncertainty window included in the considered 20 My window (Le Goff et al., 2002). For example, the pole of the APWP for 540 Ma (window 530–550 Ma) is determined using, among other poles, a pole for 520 Ma with uncertainty window 496–544 Ma, with then a weighting of (544–530)/20. The obtained APWP is presented on Fig. 17. The curve is very similar to that of McElhinny et al. (2003) but more regular because of integration of all possible data for each time window. That is in particular the case for the APWP between 400 and 500 Ma. The confidence zones are also often much smaller. The pole for 350 Ma (then including our new paleomagnetic datum) obtained from this curve (11.4° S, 29.9° E, $K=7.1$, $A_{95}=4.7°$ Fig. 17) is close (overlap of confidence zones) to the corresponding one of McElhinny et al. (2003) compilation for the same age.

7. Geodynamical implications

Using this new mean pole for Gondwana and that calculated by McElhinny and McFadden (2000) for Laurussia, a paleogeographic reconstruction of the Pangea is proposed for the beginning of the Carboniferous (Fig. 18). This paleogeographic map shows that the pre-Hercynian ocean between Gondwana and

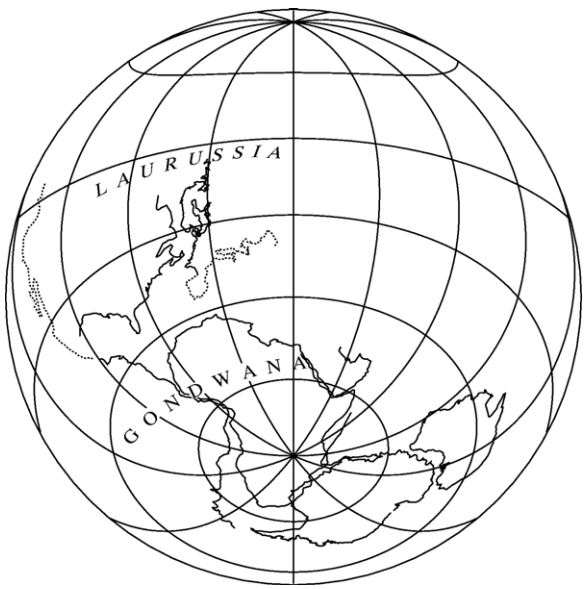


Fig. 18. Paleogeographic map Gondwana–Laurussia during the beginning of the Carboniferous.

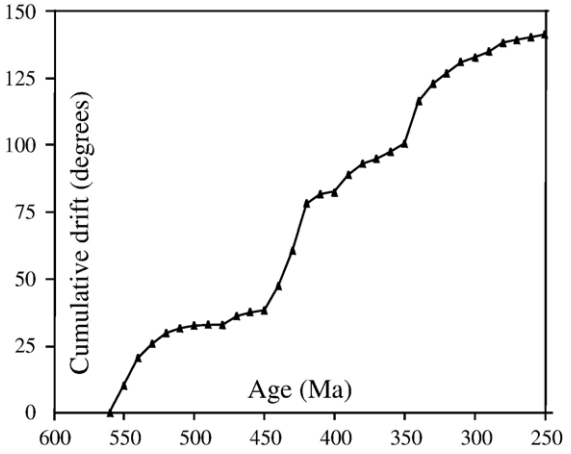


Fig. 19. Simplified cumulative latitudinal drift curve for Gondwana.

Laurussia is still not close at this time. The paleolatitude of the main continents is not very different from those obtained by Van der Voo (1993) and McElhinny et al. (2003). There are also no major differences with the paleogeographic map of Scotese and McKerrow (1990) obtained from paleoclimatic data.

To compare the drift velocity of Gondwana during Middle Paleozoic with that previously calculated by Meert et al. (1993), we computed the cumulative latitudinal movement of Gondwana from the paleolatitude drift of our sampling site by using the paleopoles data from our new APWP. The obtained results show (Fig. 19) that similarly a rapid drift episode existed around 320–350 Ma. The paleolatitudinal drift depends on site location if the continent is affected by a rotation, and that explains the difference of velocity with the Meert et al. (1993) curve.

It is interesting to underline that a relatively regular eastward drift of the African paleomagnetic pole location can be observed from Upper Devonian (Aifa et al., 1990), Tournaisian (this study), Serpukhovian–Lower Moscovian (Derder et al., 2001a) and to Upper Carboniferous (Daly and Irving, 1983; Henry et al., 1992, 1999; Derder et al., 2001b,c; Merabet et al., 1999). This regular eastward drift exists also in the new Gondwana APWP during the Upper Devonian–Lower Carboniferous (Fig. 18). This trend of the APWP is consistent with a large clockwise rotation between 370 and 340 Ma. This is in agreement with the geodynamical model proposed by Matte (1986).

8. Conclusion

To improve the knowledge of Gondwana position during the Upper Devonian–Lower Carboniferous, paleomagnetic investigations have been undertaken on dolerites emplaced as sills and dykes cropping out in the Tin Serririne area (Saharan craton, Algeria). The absolute dating gave a Tournaisian age for these dolerites. Despite the presence of magnetic overprints, the primary magnetization has been isolated in 12 sites giving a new African Lower Carboniferous paleomagnetic pole, which is coherent with the previous reliable data. This pole improves the APWP of Gondwana, for this key period of the evolution of the Pangea. This APWP confirms also the earlier paleogeographic model involving a large clockwise rotation of Gondwana during the Hercynian times.

Acknowledgements

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